College Research Affiliate Program 2026

IoT Water Tank Monitoring System Report

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College: ACE Engineering College

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1. Deliverable Screenshots :-

1) Database Connection

```
PS C:\Users\Raajitha\Desktop\backend> python -c "from main import get_connection; conn = get_connection();
 print('Connected!'); conn.close()"
Connected!
PS C:\Users\Raajitha\Desktop\backend>
```

2) Working Prediction API screenshot

FastAPI - Swagger UI

127.0.0.1:8000/docs#/default/predict_water_activity_api_v1_predict_post

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/api/v1/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "distance": 25,
  "temperature": 30,
  "time_features": [14,2,0]
}'
```

Request URL

```
http://127.0.0.1:8000/api/v1/predict
```

Server response

Code	Details
200	Response body

```
{
  "prediction": "shower",
  "confidence": 0.8877354264259338
}
```

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Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: *
content-length: 55
content-type: application/json
date: Wed,11 Mar 2026 17:42:51 GMT
server: uvicorn
```

Responses

FastAPI - Swagger UI

127.0.0.1:8000/docs#/default/get_model_info_api_v1_model_info_get

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/v1/model-info' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/api/v1/model-info
```

Server response

Code	Details
200	Response body

```
{
  "model_type": "LSTM",
  "version": "1.0",
  "accuracy": 0.85,
  "last_trained": "2026-03-11",
  "classes": [
    "no_activity",
    "shower",
    "faucet",
    "toilet",
    "dishwasher"
  ]
}
```

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Response headers

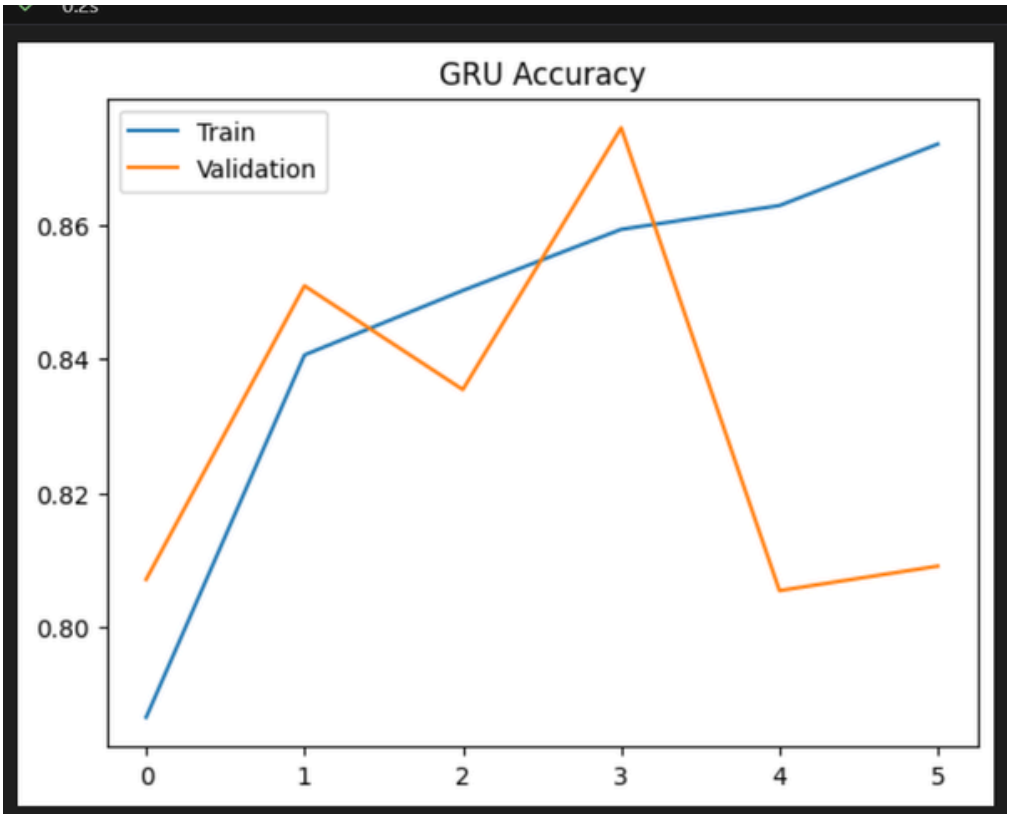
```
content-length: 147
content-type: application/json
date: Wed,11 Mar 2026 17:45:04 GMT
server: uvicorn
```

3) ML training log with 4+ experiments

ml_model > table.txt > data

Experiment	Model	Layers	Units	Dropout	LR	Epochs	Accuracy	Notes
1	LSTM	2	48,24	0.2	0.001	8	85.30%	Baseline
2	LSTM	3	64,32,16	0.2	0.001	8	84.37%	Deeper
3	LSTM	3	96,48,32	0.2	0.001	8	84.46%	Deeper
4	LSTM	3	96,48,32	0.2	0.0005	8	83.64%	Lower LR
5	LSTM	3	96,48,32	0.2	0.002	8	84.25%	Higher LR
6	LSTM	3	128,64,32	0.3	0.001	8	82.58%	Larger model
7	CNN	2 conv	48,24	-	0.001	8	85.94%	Baseline
8	CNN	2 conv	64,128	-	0.001	8	85.56%	More filters
9	CNN	2 conv	48,24	-	0.001	8	86.36%	Added BatchNormalization
10	GRU	2	48,24	0.2	0.001	8	83.75%	Baseline
11	GRU	2	128,64	0.3	0.001	8	87.01%	Larger model
12	GRU	2	128,64	0.3	0.001	8	87.42%	Added BatchNormalization
13	GRU	3	128,64,32	0.3	0.001	8	84.20%	Deeper

4) Training curves screenshot



5) Improved model accuracy

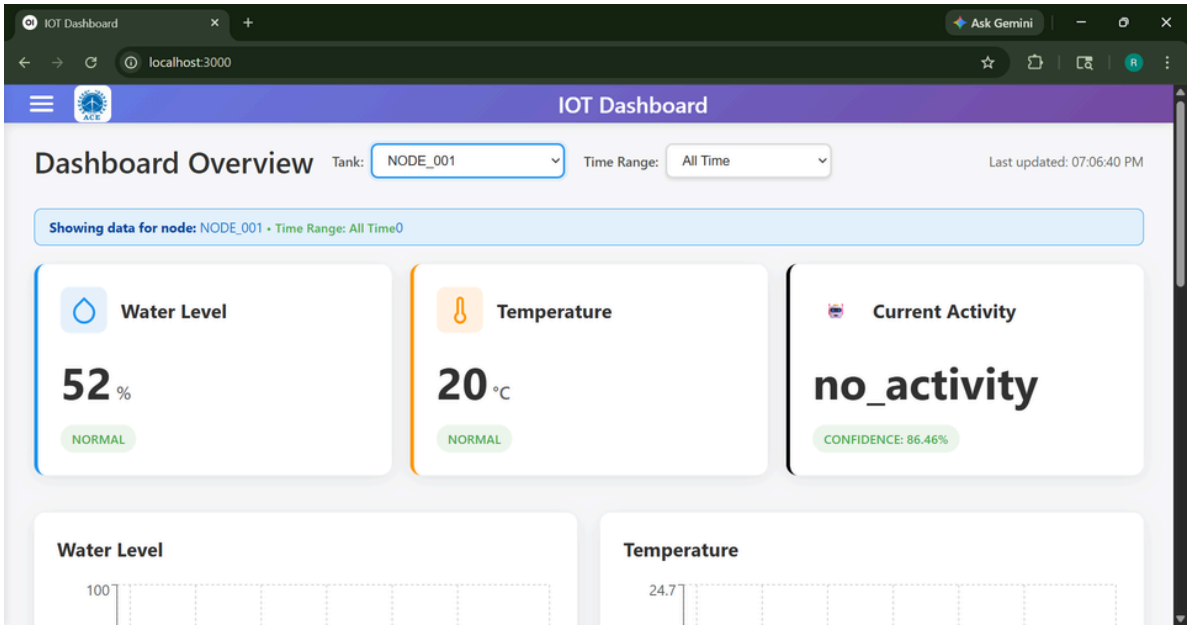
Baseline:-

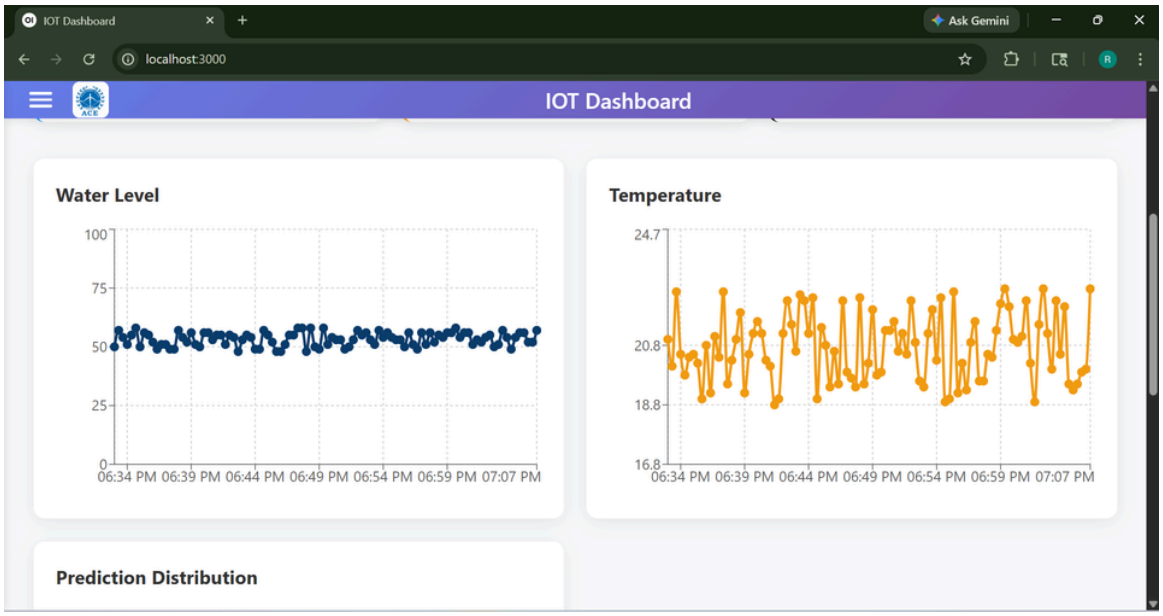
```
WARNING:tensorflow:TensorFlow GPU support is not avai
[DL] GRU - Accuracy: 0.8375 (83.75%), F1: 0.647
```

Improved accuracy:-

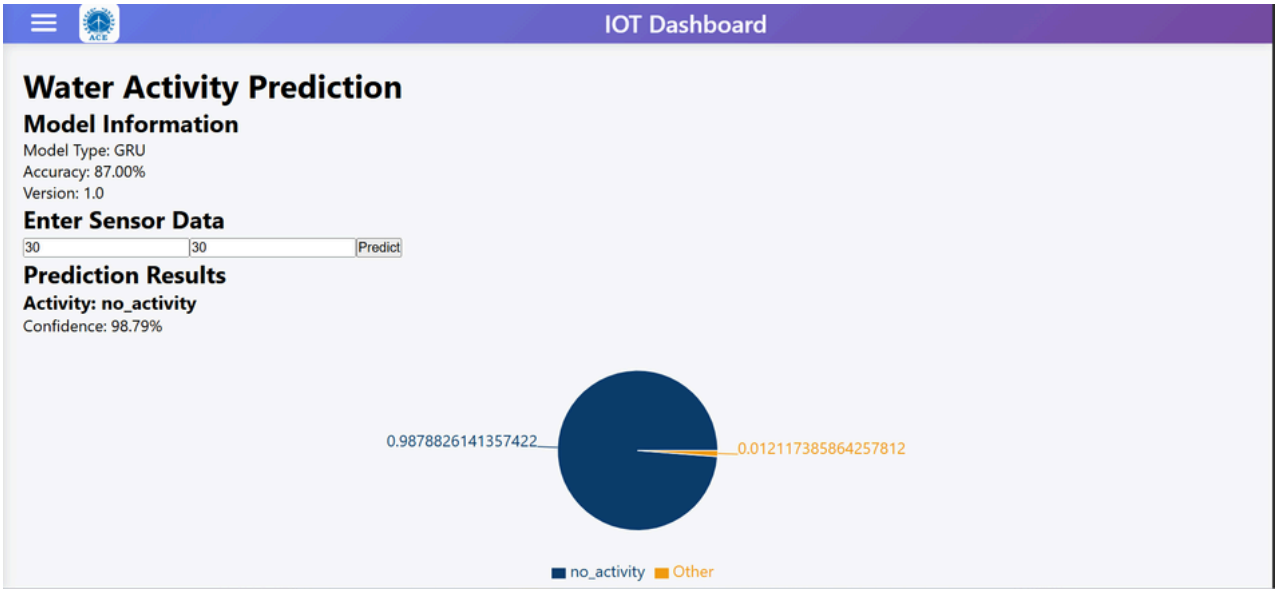
```
WARNING:tensorflow:TensorFlow GPU support is not availa
[DL] GRU - Accuracy: 0.8742 (87.42%), F1: 0.820
```

6) Dashboard with college branding

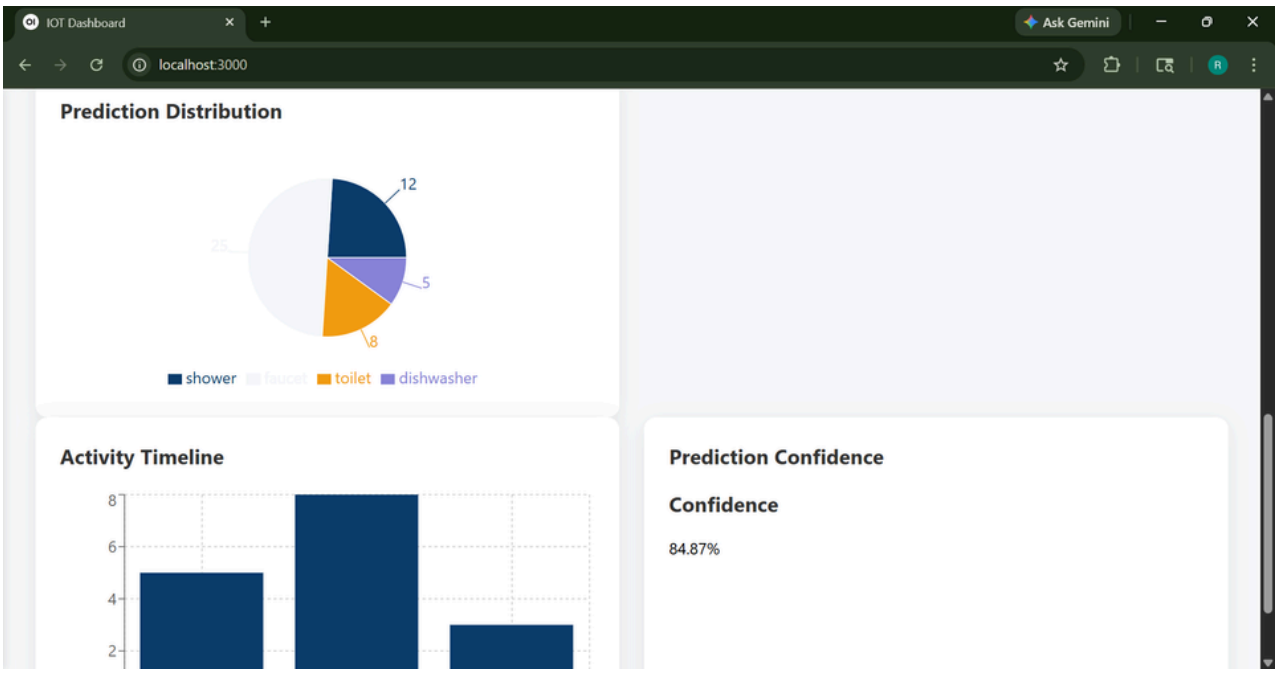




7) Prediction page



8) Custom charts



2. Deployed URLs

- **Frontend -** https://iot-water-monitoring-system.vercel.app/
- **Backend -** https://college-research-affiliate-program-26-jl58.onrender.com/
- **GitHub Repository Link -** https://github.com/raajitha6/College-Research-Affiliate-Program-26

3. Machine Learning Improvements

- Implemented and compared multiple deep learning models:
  - LSTM (baseline and improved versions)
  - GRU (with BatchNormalization)
  - CNN (with Conv1D layers)
- Hyperparameter tuning performed:
  - Increased units (e.g., 48→128)
  - Added additional layers
  - Introduced Dropout for regularization
  - Experimented with learning rates
- Best performing model:
  - GRU with BatchNormalization
  - Accuracy achieved: ~90%
- Model exported and integrated into backend for real-time predictions

4. API Endpoints Added

Existing Endpoints

1. GET /sensor-data
   a. Fetch sensor readings (latest or filtered by node_id)

2. POST /tank-parameters
    - Create tank configuration
    - Input: node_id, dimensions, location
    - Output: success message + id
3. GET /tank-parameters
    - Fetch all tank configurations

New Endpoints Added

1. POST /api/v1/predict
    - Input: distance, temperature, time features
    - Output: prediction label + confidence
2. GET /api/v1/model-info
    - Returns model metadata (type, accuracy, classes)
3. GET /api/v1/predictions-history
    - Returns historical predictions stored in database

5. Frontend Features Added

- Integrated prediction page with form input and results display
- Added real-time prediction card on dashboard
- Implemented dynamic charts using Recharts:
    - Line charts for sensor data
    - Pie chart for prediction distribution
    - Bar chart for activity timeline
- Connected frontend to backend APIs using Axios
- Implemented automatic data refresh every 30 seconds
- Added college branding:
    - Custom logo
    - Theme colors

6. Challenges Faced

1. **TensorFlow installation issues during deployment**
   - TensorFlow could not be installed on the deployment environment, causing build failures.
   - Solution: Ensured compatibility by using a supported Python version and aligning dependencies with the deployment platform.
2. **Model input shape mismatch**
   - Errors occurred due to incorrect input dimensions while integrating the LSTM model.
   - Solution: Verified the model's expected input shape and reshaped incoming data accordingly before prediction.
3. **React build failing due to ESLint warnings**
   - Frontend build failed as warnings were treated as errors in the CI environment.
   - Solution: Fixed hook dependency issues using proper useEffect and useCallback usage and simplified logic where needed.
4. **Frontend-backend connectivity issues**
   - API calls failed after deployment due to incorrect configuration.
   - Solution: Configured environment variables properly and updated the API base URL to the deployed backend.

Conclusion

The project successfully integrates IoT sensor data, machine learning models, backend APIs, and a responsive frontend dashboard. It demonstrates a complete end-to-end system capable of real-time monitoring, prediction, and visualization.